

Problem 1: Can prices decentralize an efficient assignment?

- (a) **Solution.** Let x_{ij} denote the fraction of item j assigned to buyer i . The LP is

$$\begin{aligned} \max_{x \geq 0} \quad & \sum_{i \in \mathcal{B}} \sum_{j \in \mathcal{I}} v_{ij} x_{ij} \\ \text{s.t.} \quad & \sum_{j \in \mathcal{I}} x_{ij} \leq 1 \quad \text{for every } i \in \mathcal{B}, \\ & \sum_{i \in \mathcal{B}} x_{ij} \leq 1 \quad \text{for every } j \in \mathcal{I}. \end{aligned}$$

The first constraints express unit demand and the second express unit supply. The bipartite matching polytope is integral, so an optimal extreme point has $x_{ij} \in \{0, 1\}$ even though the LP only imposes nonnegativity. Thus the LP has an optimum that is an actual assignment. $\square$

- (b) **Solution.** Associate $u_i \geq 0$ with buyer i 's constraint and $p_j \geq 0$ with item j 's constraint. The dual is

$$\begin{aligned} \min_{u, p \geq 0} \quad & \sum_{i \in \mathcal{B}} u_i + \sum_{j \in \mathcal{I}} p_j \\ \text{s.t.} \quad & u_i + p_j \geq v_{ij} \quad \text{for every } i \in \mathcal{B}, j \in \mathcal{I}. \end{aligned}$$

Equivalently, $u_i \geq v_{ij} - p_j$ for every item j . At market prices p , the right-hand side is buyer i 's net surplus from item j ; the dual variable u_i will equal the buyer's maximum attainable surplus at optimum. $\square$

- (c) **Solution.** The complementary-slackness conditions are

$$\begin{aligned} x_{ij} > 0 & \implies u_i + p_j = v_{ij}, \\ u_i > 0 & \implies \sum_{j \in \mathcal{I}} x_{ij} = 1, \\ p_j > 0 & \implies \sum_{i \in \mathcal{B}} x_{ij} = 1. \end{aligned}$$

The first condition is the key one for decentralization: on every allocated edge, value equals buyer utility plus item price. The other two conditions say that a buyer with positive utility receives an item and an item with positive price is allocated. Equivalently, an unassigned buyer has $u_i = 0$, and an unallocated item has $p_j = 0$. $\square$

- (d) **Solution.** If $x_{ij} > 0$, complementary slackness makes the corresponding dual constraint tight: $u_i + p_j = v_{ij}$, so $u_i = v_{ij} - p_j$. Dual feasibility also gives $u_i + p_k \geq v_{ik}$ for every k , hence

$$v_{ik} - p_k \leq u_i = v_{ij} - p_j.$$

Finally $u_i \geq 0$, so the assigned item weakly beats both every other item and the outside option. Therefore the efficient assignment x can be decentralized at the dual prices p : every buyer is willing to accept the item assigned to them.

If buyers are indifferent among several items, other selections from their demand correspondences may fail to form a feasible assignment. The guarantee is that the efficient assignment can be selected from the buyers' demand sets. $\square$

(e) **Solution.** The efficient assignment is

$$1 \rightarrow A, \quad 2 \rightarrow C, \quad 3 \rightarrow B,$$

with welfare $10 + 7 + 9 = 26$. One optimal dual solution is

$$(p_A, p_B, p_C) = (5, 4, 3), \quad (u_1, u_2, u_3) = (5, 4, 5).$$

It is dual feasible and has objective

$$5 + 4 + 5 + 5 + 4 + 3 = 26,$$

equal to the primal value. The net-surplus matrix is

	A	B	C
1	5	4	0
2	4	1	4
3	-1	5	5

so buyer 1 chooses A , buyer 2 may choose A or C , and buyer 3 may choose B or C . The efficient allocation chooses A, C, B from these demand sets. Ties are allowed: complementary slackness requires the allocated item to be weakly utility-maximizing, not uniquely optimal. $\square$

(f) **Solution.** At prices $(5.8, 4, 3.7)$, net surpluses are

	A	B	C
1	4.2	4	-0.7
2	3.2	1	3.3
3	-1.8	5	4.3

Thus buyers 1, 2, 3 strictly prefer A, C, B , respectively.

At prices $(5.5, 4, 3.4)$, net surpluses are

	A	B	C
1	4.5	4	-0.4
2	3.5	1	3.6
3	-1.5	5	4.6

so the same assignment is strictly supported. Seller revenue is $5.8 + 3.7 + 4 = 13.5$ under the first vector and $5.5 + 3.4 + 4 = 12.9$ under the second. Thus supporting prices need not be unique, and supporting an efficient allocation can generate different seller revenues. Efficiency alone therefore does not select the seller's preferred supporting price vector. Buyer utilities sum to 12.5 and 13.1, respectively; in both cases utility plus revenue equals welfare 26.

Together, parts (e)–(f) illustrate two distinct facts: complementary slackness only requires the allocated item to be weakly utility-maximizing, and supporting prices need not be unique. Some optimal dual prices generate ties, while other supporting prices can separate the assigned items strictly. $\square$

Problem 2: How does a distribution of values generate demand?

- (a) **Solution.** Each $X_i(p)$ is Bernoulli with success probability $q(p) = 1 - F(p)$. Independence gives

$$Q_n(p) \sim \text{Binomial}(n, q(p)), \quad d(p) = \mathbb{E}[Q_n(p)] = nq(p), \quad \text{Var}(Q_n(p)) = nq(p)(1 - q(p)).$$

Therefore

$$\text{sd}(Q_n(p)) = \sqrt{nq(p)(1 - q(p))} = \sqrt{nF(p)(1 - F(p))}.$$

When $q(p) > 0$, its relative standard deviation is

$$\frac{\sqrt{\text{Var}(Q_n(p))}}{\mathbb{E}[Q_n(p)]} = \sqrt{\frac{1 - q(p)}{nq(p)}} = \sqrt{\frac{F(p)}{n(1 - F(p))}}.$$

For fixed $q(p) \in (0, 1)$,

$$d(p) = \Theta(n), \quad \text{sd}(Q_n(p)) = \Theta(\sqrt{n}),$$

so relative fluctuations are of order $n^{-1/2}$. □

- (b) **Solution.** Random total revenue is $R_n(p) = pQ_n(p)$, so $r_n(p) = \mathbb{E}[R_n(p)] = np(1 - F(p))$. The per-buyer expected revenue curve is

$$r(p) := \frac{r_n(p)}{n} = p(1 - F(p)).$$

For $V \sim \text{Uniform}[0, m]$ and $p \in [0, m]$,

$$r_n(p) = np \left(1 - \frac{p}{m}\right), \quad r'_n(p) = n \left(1 - \frac{2p}{m}\right), \quad r''_n(p) = -\frac{2n}{m}.$$

Thus $p^* = m/2$, and revenue is strictly concave on $[0, m]$.

For $V \sim \text{Exp}(\lambda)$,

$$r_n(p) = npe^{-\lambda p}, \quad r'_n(p) = ne^{-\lambda p}(1 - \lambda p),$$

so $p^* = 1/\lambda$. Moreover,

$$r''_n(p) = n\lambda e^{-\lambda p}(\lambda p - 2),$$

which changes sign at $2/\lambda$. Revenue is not globally concave in price. □

- (c) **Solution.** The functions $r(p) = p(1 - F(p))$ and $r(q) = qF^{-1}(1 - q)$ describe the same per-buyer expected revenue in price and quantile coordinates, respectively. For the uniform distribution,

$$p(q) = m(1 - q), \quad r(q) = mq(1 - q), \quad r''(q) = -2m < 0.$$

For the exponential distribution,

$$p(q) = -\frac{1}{\lambda} \log q, \quad r(q) = -\frac{q}{\lambda} \log q, \quad r''(q) = -\frac{1}{\lambda q} < 0.$$

Thus both curves are concave in quantile space, although exponential revenue is not globally concave in price space. □

(d) **Solution.** Write $v = p(q) = F^{-1}(1 - q)$. Since $q = 1 - F(v)$, $dv/dq = -1/f(v)$. Therefore

$$r'(q) = \frac{d}{dq}(qv) = v + q \frac{dv}{dq} = v - \frac{q}{f(v)} = v - \frac{1 - F(v)}{f(v)} = \phi(v) = \phi(p(q)).$$

As q increases, $p(q)$ decreases. Hence $r'(q) = \phi(p(q))$ is nonincreasing in q exactly when $\phi(v)$ is nondecreasing in v . This is concavity of r and regularity of the distribution. $\square$

(e) **Solution.** Because $1/h(v) = (1 - F(v))/f(v)$,

$$\phi(v) = v - \frac{1}{h(v)}.$$

If h is nondecreasing, then $1/h$ is nonincreasing; consequently $v - 1/h(v)$ is nondecreasing. Thus MHR implies regularity. When h is differentiable and positive, $\phi'(v) = 1 + h'(v)/h(v)^2 \geq 1$. $\square$

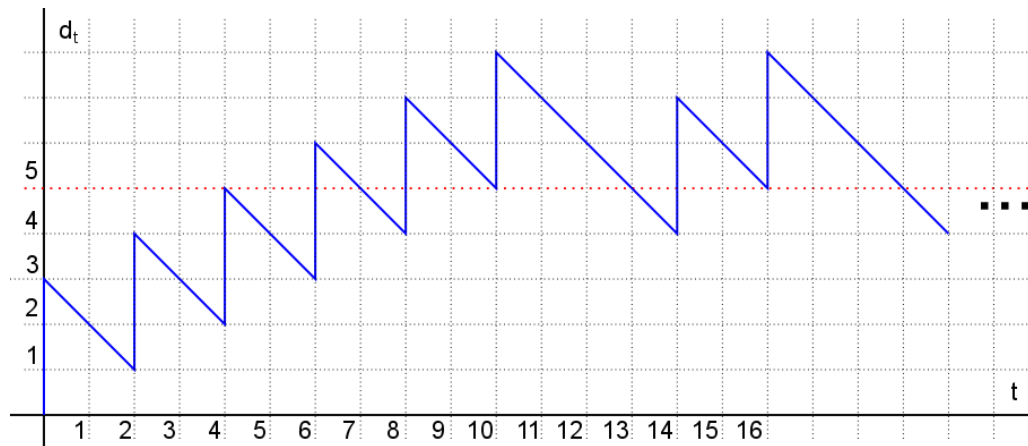
Problem 3: Pricing a congested service

(a) **Solution.** When $w = 0$, the largest acceptable access price is 5. More generally it is $5 - w$ when $0 \leq w \leq 5$. If $w > 5$, delay cost already exceeds the robot's mission value, so no nonnegative price induces it to dock. $\square$

(b) **Solution.** The full-extraction policy charges $\tau(w) = 5 - w$ for $w \leq 5$ and lets the robot dock; the robot reroutes when $w > 5$. Starting from zero, the workload evolves as

$$0, 1, 2, 3, 4, 5, 6, 4, 5, 6, \dots$$

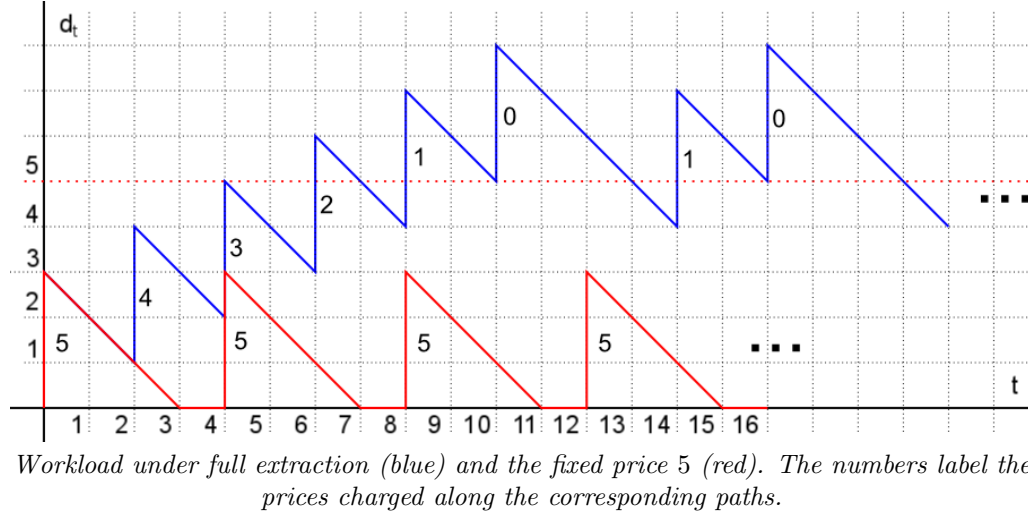
The eventual three-arrival cycle earns 1 at $w = 4$, 0 at $w = 5$, and 0 when the robot reroutes at $w = 6$. Arrivals are two cycles apart, so the cycle lasts six cycles. Long-run average revenue is $1/6$ credit per cycle.



Full extraction creates a ratchet: accepted reservations push the workload upward until it cycles near the docking boundary $w = 5$.

$\square$

- (c) **Solution.** At access price 5, only a robot arriving at $w = 0$ docks. The workload cycle is $0, 1, 0, 1, \dots$. One repair generating 5 operator credits occurs every two arrivals, or every four cycles. Long-run average revenue is $5/4$ credits per cycle.



□

- (d) **Solution.** At $w = 0$ the robot docks at price 5; at $w = 1$ the next robot docks at price 4; at $w = 2$ a price of 4 is too high, so that robot reroutes. The workload cycle is $0, 1, 2, 0, 1, 2, \dots$. The policy earns $5+4 = 9$ operator credits every three arrivals, or every six cycles. Its long-run average revenue is $9/6 = 3/2$ credits per cycle. □

- (e) **Solution.** The average revenues are $1/6$, $5/4$, and $3/2$. Full extraction admits robots whose delay makes current willingness to pay very small; those repairs consume future capacity and create still more delay. The scarce resource is repair-bay availability over time. An access price therefore rations future capacity and represents an opportunity cost.

If payments are transfers, a robot that docks at workload w contributes $5 - w$ to city surplus. A policy must account for the state it creates for future arrivals, not only revenue available from the current arrival. □